Ex: $\mathbb{G}_m \curvearrowright \mathbb{G}_m$: $\quad \underset{G}{\overset{\shortparallel}{\mathbb{G}_m}} \curvearrowright \underset{X}{\overset{\shortparallel}{\mathbb{G}_m}}$

weak: $DMod(X/G)$

strong: $DMod(X/G)$

Exercise: $\mathbb{G}_m$-module/rep $\longleftrightarrow$ $\mathcal{O}(\mathbb{G}_m)$-comodule

$v \longmapsto \underbrace{v_1 \otimes z^{n_1}}_{\text{weight } n_1} + \cdots + \underbrace{v_r \otimes z^{n_r}}_{\text{eigenvectors for character } z \mapsto z^{n_r}}$

alg. cl.?

- $\mathbb{G}_m$-eq sh on $\mathbb{G}_m$ $\longleftrightarrow$ vector spaces V } sheaf

$V[z, z^{-1}]$ $\quad$ $wt(z) = 1$ $\quad$ $wt = 0$

V

- D-module structure; compatible grading

$wt(\partial) = -1$

$wt(1) = 0$

$D(\mathbb{G}_m) = \dfrac{k\langle z, \partial_z, z^{-1}\rangle}{[\partial_z, z] = 1}$

D-module action on $V[z, z^{-1}]$ $\xrightarrow{wt\ 0}$ $\dfrac{k\langle \partial_z, z\partial\rangle}{\partial z - z\partial = 1} \curvearrowright V$ $\longleftrightarrow$ action of $z\partial$ on V

generates $\mathbb{G}_m$ action

V

matrix

log of monodromy

---

strong: extra condition: α_x come from equivariance

$x \in \mathfrak{g}$ $\quad$ $\overset{\shortparallel}{\delta_x}$ come from diff. op.

$x \in \mathfrak{g} = Lie(\mathbb{G}_m)$ $\quad$ $\alpha_x = 0$ $\quad \Rightarrow \quad$ matrix $= 0$. i.e.

$\downarrow$

$z\partial \in D(\mathbb{G}_m)$ $\quad$ $\delta_x = $ matrix $\quad$ strong D-module are just V

weak

equivariance: vector bundle

$+$

D-module str.: monodromy

strong:

equiv, monodromy $= 0$.

Exercise: $DMod(\mathbb{A}^1/\mathbb{G}_m)$

Example: G discrete.

$DMod(X/G) = DMod(X/G)$ $\quad \|$

Facts: ① $G \curvearrowright X$ w/ connected centralizers,

$DMod(X/G) \longrightarrow DMod(X)$ fully faithful. i.e. strong G-eq. is property not structure

$\subset$ $\quad$ $\subset$

FAILS if derived structure b/c things have cohomology. $\longrightarrow$ cohomology category

image: singular supp conical to orbits

② If G connected unipotent, then orbits have no cohomology.

centralizers connected

$\Rightarrow$ the converse holds. $G \curvearrowright X$ free + proper

③ If $X \to X/G$ is geometric quotient, then $DMod(X/G)$ $\quad$ strong eq. G-D-mod on X

(e.g. G-torsor)

$\overset{\shortparallel}{DMod(X/G)}$ $\quad$ $DMod$ on X/G

Twisted Beilinson-Bernstein:

① $W \curvearrowright \mathfrak{g}^*$ → acts normally, $\mathfrak{g} \xrightarrow{\pm P} \mathfrak{g}^*$

→ $\boxed{\text{shift the } W\text{-action}}$

② BWB

$$\begin{tikzcd} \mathrm{Coh}^G(G/B) = \mathrm{Rep}(B) \arrow[d, "\text{induction}"] & \mathrm{Rep}(H) \arrow[l, "\text{inflate}"'] \\ \mathrm{Coh}^G(\mathfrak{a}) = \mathrm{Rep}(G) & \end{tikzcd}$$

$H = B/[B,B]$.

$\mathrm{Rep}(H) \ni$ character $\lambda \in X^*(H)$

Question: positive or negative?

$\boxed{\text{Chosen Borel}}$

- Usual convention: insist that $\mathfrak{n} \subseteq \mathfrak{b}$ positive weights

$\Rightarrow$ λ positive $\rightsquigarrow$ $\mathcal{L}_\lambda$ is ample iff $\lambda \ll 0$ — Check

$\rightsquigarrow$ $\Gamma(\mathcal{L}_\lambda) = V_{-\lambda}$ for $\lambda \ll 0$

- $\boxed{\mathfrak{n} \subseteq \mathfrak{b} \text{ negative weights}}$

Def: T is an algebraic torus, $p: \tilde{X} \to X$, T-torsor

- $\mathrm{DMod}(\tilde{X}; T) \simeq (p_* \mathcal{D}_{\tilde{X}})^T$-module on X (relative version of exactly)

- $(p_* \mathcal{D}_{\tilde{X}})^T \simeq \mathcal{D}_X \otimes_k \mathrm{Sym}_k^{\mathfrak{t}}$ ← monodromy operators

Def $\boxed{\mathcal{D}_{G/B,\lambda} = p_*(\mathcal{D}_{G/U})^H \otimes_{\mathrm{Sym}} k_\lambda}$ $\lambda \in \mathfrak{t}^*$

sheaf of twisted diff. op. on G/B.

If λ integral, this arises via $\mathrm{End}(\mathcal{L}_\lambda)$

Thm Assume $\lambda \in \mathfrak{g}^*$ $\boxed{\langle \lambda, \alpha^\vee \rangle \neq -1, -2, -3, \ldots, \quad \alpha \in \Delta^+}$ ← includes all dominant weights.

then, $\Gamma: \mathrm{DMod}(G/B, \lambda) \xrightarrow{\simeq} \mathrm{Rep}(\mathfrak{g})_{[\lambda]}$

① Every central ch. has a representation

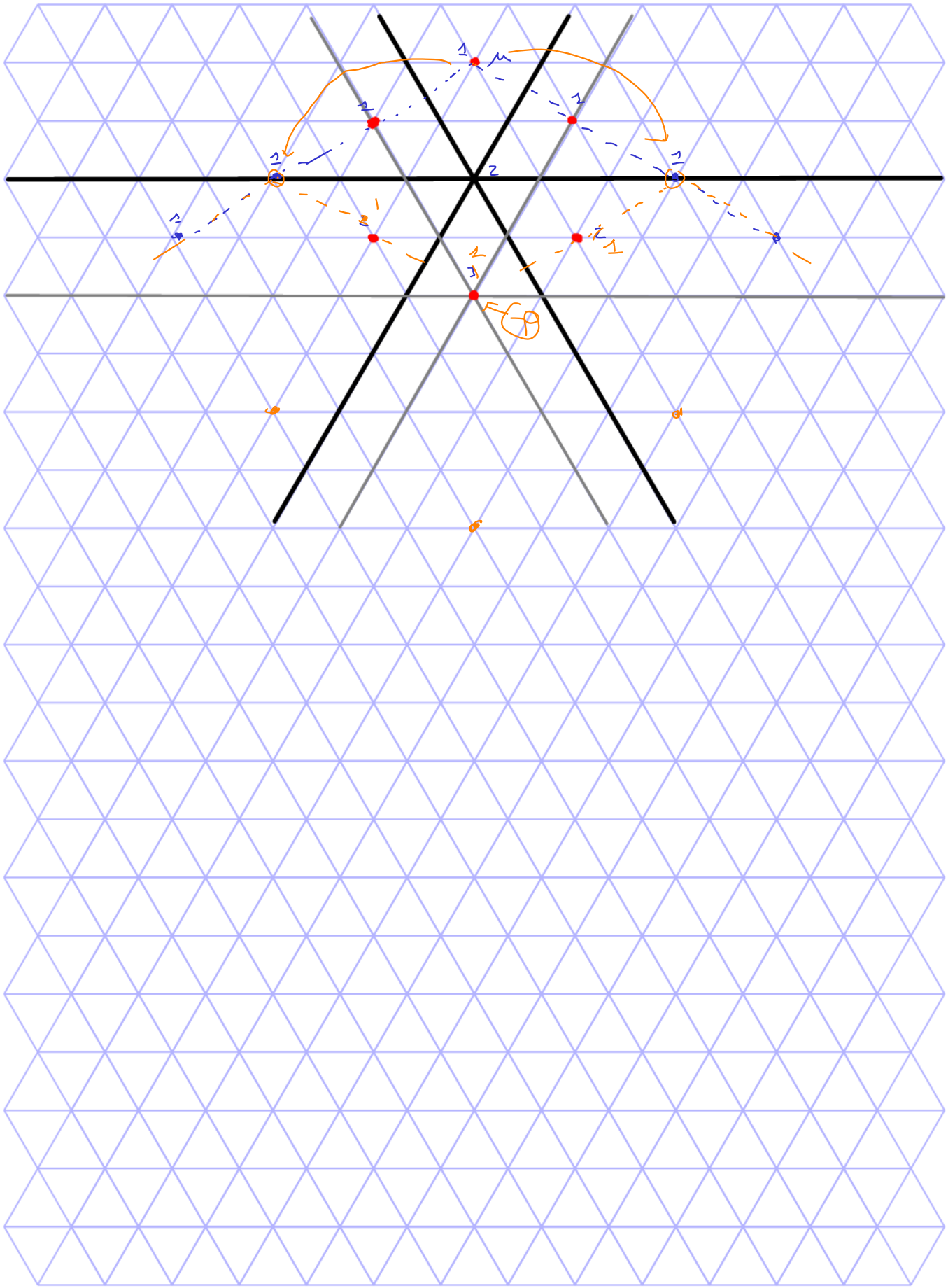
② condition excludes
- integral $\Rightarrow$ has to be dominant
- anti dominant $\Rightarrow$ has to be generic, i.e. $\mathrm{DMod}(N \backslash G/B, \lambda)$ completely non-integral

$\mathrm{DMod}^N(G/B, \lambda) \simeq \mathcal{O}'_{[\lambda]}$

$M(\mu) \to L(\mu)$

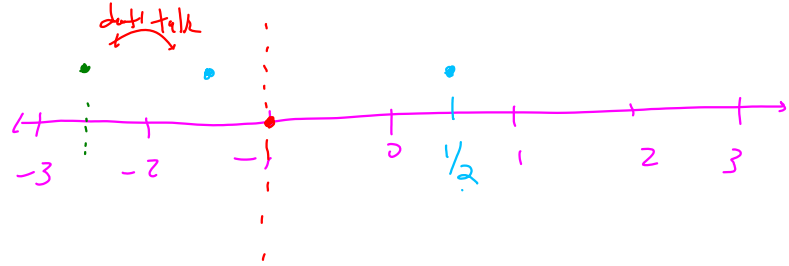
S_k

reflection hyperplanes



Free Triangle Graph Paper from http://incompetech.com/graphpaper/triangle/

$\underline{Sl_2}$:



$M(1/2)$

$M(-5/2)$

different centers rat lattice.

at $\underline{\text{integral}}$ $\lambda = 0$ weights: $L(0)$, $L(-2)$

at $\underline{\text{nonintegral weights}}$: $\mathcal{O}_{[\lambda]}$ can decompose further

$L(0) = M(0) - M(-2)$

at singular weight $\mathcal{O}_{-1} = L(-1) = M(-1)$

$\underline{\text{Next time}}$: D-modules, derived cats / $\mathcal{D}$ fns

$D(\mathcal{O}_{!}/G_m) \longrightarrow D(G_m)$

$\mathrm{Rep}(G_m) \longrightarrow \mathrm{Vect}$.